MotoMedic answer architecture decisions

Companion note to TRD v0.5 11 October 2026

We will retain A and B for the application and deliver A through verified streaming. C will remain an experiment outside the current UI path. This note explains the owner decision, the design choices considered and the evidence used, so a reader can distinguish tested decisions from options that still need evaluation.

Selected answer methods

Choice	What it does	Decision and reason
A component selection	Gemma selects indices into exact reviewed wording; server supplies the approved text	Selected with streaming. All three cases passed in both passes; exact candidate reuse avoided generation on repeats.
B short generation	Gemma writes concise paraphrases, followed by a separate claim verifier	Retained as a methodology. All three cases passed in both passes; supports wording flexibility while retaining verification.
C verified blocks	Generation streams independent blocks, each released after verification	Not selected for the UI. Clock failed in both passes; successful paced response times were high and earlier safe output was not demonstrated.

A removes the separate online semantic-verifier call only because the output is selected from exact approved wording. It retains source applicability, approved dependency completeness, warnings, prerequisites, typed session eligibility, citation checks and durable session save. A is not an unrestricted generator without verification. B retains its semantic verifier. No model score can authorize a new repair instruction.

A streaming will emit the whole verified answer unit after its gates and session save. Provider token fragments and partial warnings remain withheld. This is a delivery contract; it does not promise that repair text appears during selection. C is not an automatic fallback. These choices are prospective UI methods, not a claim that the public app is deployed.

Measured comparison and its limits

The latest hosted experiment used Gemma 26B gemma-4-26b-a4b-it with minimal thinking, the same manual/index and three cases: GEN003 mirrors, EXP003 signal cancellation and GEN026 clock settings. Nine empty-cache turns and nine immediate repeats returned 8/9 SUPPORTED in each pass. All A/B cases passed; C clock failed twice. There were 27 attempts, zero retries, 26 known HTTP200 results and one unknown HTTP status.

Method case	Retrieval	TTFT	TTFO	Full	Pacing	Full minus pacing
A Mirrors	1.71	0.66	2.88	2.88	0.00	2.88
A Signal	0.84	0.62	9.54	9.57	7.69	1.88
A Clock	0.86	0.61	9.69	9.69	7.88	1.80
B Mirrors	1.71	0.67	18.13	18.13	13.51	4.62
B Signal	0.84	0.70	49.98	50.01	46.25	3.76
B Clock	0.86	0.85	39.12	39.12	35.55	3.57
C Mirrors	1.70	0.69	19.81	19.82	14.98	4.84
C Signal	0.84	0.69	49.93	49.96	45.84	4.12
C Clock	0.87	0.71	--	--	6.78	--

All values are seconds. Retrieval includes assembly and conditional reranking. TTFT above begins at provider transport start and represents the first non-thinking text fragment, not safe advice. TTFO and full completion are actual localhost HTTP/SSE client receipt times from request start, including setup, retrieval and pacing. Request-relative TTFT is retained in the full JSON report. Public network and browser rendering are unmeasured.

Full minus pacing is arithmetic, not an observed unpaced run. Idle waits are counted by interval union, with concurrent active provider work excluded. Successful adjusted full times were approximately 1.80-4.84 seconds. Conservative pacing was an experimental admission policy, not a verified account quota; most long observed waits were this pacing.

C clock had no delivered answer. Initial client error time was 9.63 seconds; its exact transport/gate cause and HTTP status were not retained. The repeat failed at 49.16 seconds because both verifier payloads used a Boolean array instead of the required scalar unsupported_additions field. Invalid schema is a failed verification operation, not proof that the wording was unsupported.

Latency choices that were tested

Design choice	Evidence from this run	Decision or interpretation
Sequential or concurrent dense and BM25	Three warm paired controls returned identical allowed envelopes. Concurrent first-stage time was 0.424-3.341 ms slower.	Concurrency remains implemented; no demonstrated latency benefit. BM25 took 0.30-0.70 ms, so embedding and reranking are larger opportunities.
Candidate cache or bypass	All three A repeats hit; actual full times were 2.03, 1.24 and 1.18 seconds. B/C paraphrases missed.	Keep exact reviewed candidate reuse. Similarity is lookup only; scope/version/state bindings and fresh gates remain mandatory. Do not claim general semantic answer caching.
Buffered or verified streaming	Mirrors and signal each required a complete safe unit. Only clock had two independent blocks, and C clock failed.	Select A streaming by owner decision. This run does not prove earlier TTFO from independent block streaming.
Cold or warm process	Fresh worker initialization was 63.48 seconds, including 6.84 seconds of controls. Readiness excluding controls was 56.64 seconds.	Investigate packaging, warm reuse, minimum instances and startup CPU boost. No hosted configuration change has been made.

Dense model/index loading consumed 38.25 seconds of startup, registry validation 2.88 seconds, reranker load 0.28 seconds and query warmups about 14.43 seconds. The separate pretrial cooldown was 60.00 seconds. GitHub installation and job setup are additional engineering overhead, not rider request latency. These measurements must not be relabeled as Cloud Run cold-request times.

The deployed Qdrant service separately restores stored immutable vectors on startup. It does not parse or re-embed the manual per request. CPU query-model loading in the benchmark worker and Qdrant restoration are distinct cold-start paths; each needs its own measurement.

This three-case engineering comparison supports the current bounded owner decision. It does not establish population p95, held-out safety performance, a universal speed ranking or field repair success. Reversing the choice requires comparable reviewed cases, clear failure accounting and unchanged evidence gates.

Broader design choices considered

The design work also considered the alternatives below. The earlier decision register in 04_To_Build_Evaluation_Decisions.docx and HOSTED_INDEX_DECISION.md contains the detailed history. Options marked considered were not all implemented or tested; the distinction prevents an alternatives list from implying evidence that does not exist.

Area	Current design and alternatives	Evidence status
Source and parsing	Immutable owner manual; Docling with flagged page 103 fallback. Other layout parsers or Document AI considered.	Hosted ingestion completed; no approved comparative parser winner.
Chunking	1,024-token parent-child chunks; flat chunks, late chunking and alternate procedure boundaries considered.	Selected candidate has 532 children and 277 parents; no held-out chunking superiority claim.
Embedding	BGE M3 selected; MiniLM compared.	Draft hybrid cited-page coverage was BGE 23/27 versus MiniLM 21/27; in-sample diagnostic only.
Retrieval	Dense plus BM25 with RRF; dense-only and sparse-only alternatives.	Hybrid implemented; exact relevance scores remain candidate signals, not support proof.
Graph	Lightweight verified provenance links; full GraphRAG and wider hops considered.	Only verified dependencies can authorize a step; broad graph benefit remains unproven.
Reranking	Conditional cross-encoder; off, always-diagnostic and late interaction alternatives.	Conditional policy owner approved. Earlier BGE page coverage 23/27 without versus 22/27 with reranking; calibration remains scope-bound.
Controller	Deterministic typed state and TRUE/FALSE/UNKNOWN eligibility; Decisions API or Jev considered.	Models may interpret or rank eligible options, never replace controller authority.
Grounding	Exact reviewed wording in A or composition plus verification in B; independent block verification in C.	A/B selected after measured experiment; post facto A audit remains a UI prerequisite.
Photo and voice	Visible observations only; visual-page retrieval or ColPali considered. Optional STT, including Sarvam, considered.	General vision/voice performance not established by this text benchmark.
Hosting and traces	Private hosted build, Cloud Run Qdrant and Storage; Streamlit UI. Managed vector services, other UI hosts and Langfuse considered.	Private retrieval deployment exists; public UI, cost and end-to-end delivery checks are separate work.

Next work before UI integration

The owner requested a new chat to complete two concrete tasks. The durable handoff is TRD_UI_FOLLOWUP_HANDOFF.md. Do not rerun composition merely to test whether the old output passes verification, and do not enable C in the UI.

Post facto semantic audit of A

Recover the exact A wording, claim order, citations and scope from hosted run 38122505406 and its approved registry bindings. Run the removed verification step against the original cited source and required reviewed meaning for mirrors, signal and clock. Retain initial and repeat provenance; identical repeat wording may share one verifier result with an explicit exact hash match. The result is pending, not assumed to pass.

Record each claim as validly accepted, validly rejected or unassessed because of malformed schema/provider failure. Keep conditions, warning completeness, unsupported additions and citation support explicit. Use a fresh bounded hosted audit protocol, budget check and pacing ledger rather than reusing the exhausted v4 benchmark authorization. Keep Gemma 26B unless the owner changes it; never relax the scalar Boolean schema to hide a failure.

Minimum instance cost and startup CPU boost

Investigate current official Cloud Run prices for us-central1 and the actual service CPU, memory, billing mode and minimum-instance setting. Show zero versus one minimum instance under stated idle/traffic assumptions, including monthly and daily cost, free-tier/account dependence, INR assumptions and impact on the INR800 lifetime budget and INR640 billing stop. Distinguish the Qdrant service from any future query-model/API service.

Read the live startup CPU boost setting and startup path, check official pricing and propose a repeatable cold/warm experiment. Packaging pinned model assets, loading once and reusing a process are candidate improvements; verify the relevant service before attributing the measured 38.25-second worker load to Cloud Run. Research authorizes no deployment, minimum-instance, CPU, billing or public source-display change.

Evidence record

Hosted run: https://github.com/Maneshwar-Kumar/motomedic-ai/actions/runs/38122505406
Reviewed runtime commit: e910c86d2fb8a0a2f955bbea4fa4cbedbe64d82d
Raw results: reports/architecture_run_38122505406/architecture_comparison.json
Timing summary: reports/architecture_owner_measurement_summary_v4.json
Independent audit: reports/architecture_result_independent_audit_v4.json
Raw SHA256: bfd7603c1dccb84d99613c8de647de9475da4f9675d75ccdfb583c915583ac83

Earlier v3 run 38080802638 and the original comparison reports remain preserved. The v4 independent audit confirms all 18 cells, provider accounting and client timing arithmetic; it does not certify broad rider release. Current minimum-instance and CPU-boost reference pages are https://docs.cloud.google.com/run/docs/configuring/min-instances and https://docs.cloud.google.com/run/docs/configuring/services/cpu. Pricing will be checked afresh in the follow-up.